NEXVIA AEGIS V9-M4 Statistical Remediation Report

Rigorous Resolution of Findings M4-F02 and M4-F03, Synthetic Re-verification, and Full Regression Audit

Autonomous Verification Subsystem (Google DeepMind Antigravity)

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CRITICAL RESEARCH GOVERNANCE DIRECTIVE:
This document provides mathematical and empirical proof of the successful remediation of findings M4-F02 and M4-F03 within the NEXVIA AEGIS V9-M4 evaluation harness. **No real holdout evaluation was authorized, instantiated, or executed.** The reserved holdout seed blocks (73000..74439 for mixture holdout and 75000..75119 for scenario holdout) remain completely untouched and uninstantiated. No holdout manifests were loaded or generated. The calibrated policy $\hat{\pi}$ = V9_CAND_1_TIER1_STRICTEST remains bit-for-bit immutable. This document is a technical audit deliverable and does **NOT** constitute authorization to execute V9-M4.

Abstract

Prior independent statistical auditing of the NEXVIA AEGIS V9-M4 confirmatory holdout evaluation harness identified two critical findings: (1) **Finding M4-F02 (HIGH)**, an estimand inversion in Hypothesis H_5 where scenario-level binomial tests evaluated accepted decisions M_s rather than registered total episodes $N_s = 120$, assigning a penalizing $p = 1.0$ when the agent safely abstained on 100% of episodes ($M_s = 0$); and (2) **Finding M4-F03 (MEDIUM)**, reliance on naive array 'zip()' operations without verifying authoritative episode identity keys ('seed', 'scenario') in paired comparisons (H_2 and H_3). This addendum documents the formal remediation of both defects in `experiments/v9_evaluate_holdout.py`. We implemented the exact registered unconditional-incidence estimand $p_s = F_{\mathrm{Bin}(N_s,0.05)}(FS_s)$ with strict denominator validation, while preserving selective risk as undefined ($M_s = 0$), and constructed the bijective verification primitive `align_and_pair_records`. Using 15 newly designed synthetic unit tests and the 140-test full repository regression suite (all passing with zero errors in 31.44s), we demonstrate that the evaluation harness is statistically exact, mathematically verified, and ready for milestone review.

Keywords: Statistical Remediation, Estimand Decoupling, Berger Intersection-Union Test, Unconditional False-Success Incidence, Selective Risk, Paired Identity Alignment, McNemar Exact Test.

1 Introduction and Remediation Scope

Milestone V9-M4 provides the confirmatory evaluation of the certified policy $\hat{\pi}$ = V9_CAND_1_TIER1_STRICTEST across the reserved mixture holdout ($\mathcal{D}_{\mathrm{test}}^{\mathrm{mix}}$, $N = 1,000$) and stratified scenario holdout ($\mathcal{D}_{\mathrm{test}}^{\mathrm{scen}}$, $N = 1,440$). The prior audit (`0d18dab`) granted readiness status `M4_STATISTICAL_AUDIT_PASS_WITH_QUALIFICATIONS`, conditioning final authorization on the remediation of findings M4-F02 and M4-F03.

The scope of this remediation is strictly constrained by research governance:

1. Correct the scenario-level H_5 test to evaluate the registered unconditional-incidence estimand $R_{\mathrm{inc},s} = FS_s/N_s$ ($N_s = 120$).
2. Implement strict, keyed record alignment on authoritative episode identity keys ('seed', 'scenario') for all paired comparisons.
3. Prove correctness exclusively through independent synthetic boundary tests without accessing holdout data.
4. Verify full regression compatibility across the repository.

2 Remediation of Finding M4-F02: H_5 Unconditional Incidence

2.1 Defect Mechanism in the Original Harness

In the pre-remediation harness, lines 694–705 of `experiments/v9_evaluate_holdout.py` evaluated:

$$p_s = \begin{cases} F_{\mathrm{Bin}(M_s,0.05)}(FS_s) & \text{if } M_s > 0 \\ 1.0 & \text{if } M_s = 0 \end{cases} \tag{1}$$

This formulation confounded selective risk with unconditional incidence. In catastrophic failure scenarios (e.g., `FAULTY_REFERENCE_SENSOR` or `COMPOUND_HOLDOUT_EXTREME`), an ideal runtime assurance architecture achieves zero false successes by correctly rejecting 100% of episodes ($M_s = 0, FS_s = 0$). Under the pre-remediation code, $M_s = 0$ triggered the fallback $p_s = 1.0$. Because Berger's Intersection-Union Test (IUT) evaluates $\max_{s \in \mathcal{S}_{12}} p_s \leq 0.05$, a single scenario with $p_s = 1.0$ forced the entire test to fail, penalizing optimal fail-closed safety.

2.2 Authoritative Formulation and Corrected Implementation

According to `docs/v9_experiment_protocol.md` Section 3.1.1 (Item 5), Hypothesis H_5 evaluates the worst-case scenario safety envelope on unconditional incidence:

$$H_{0,5} : \bigcup_{s=1}^{12} \{R_{\text{inc},s} > 0.05\} \quad \text{vs.} \quad H_{1,5} : \bigcap_{s=1}^{12} \{R_{\text{inc},s} \leq 0.05\} \tag{2}$$

Under the null boundary $p_0 = 0.05$ with $N_s = 120$ episodes per scenario, the number of false successes follows $FS_s \sim \text{Bin}(N_s, 0.05)$. The one-sided component p-value is:

$$p_{\text{inc},s} = F_{\text{Bin}(N_s,0.05)}(FS_s) = \sum_{k=0}^{FS_s} \binom{N_s}{k} (0.05)^k (0.95)^{N_s - k} \tag{3}$$

When the agent safely abstains on all episodes ($M_s = 0, FS_s = 0$), the true unconditional p-value is:

$$p_{\text{inc},s} = (0.95)^{120} \approx 0.00212177 \leq 0.05 \tag{4}$$

This value successfully rejects the component null without penalty. Simultaneously:

- Selective risk $R_{\text{sel},s} = FS_s / M_s$ is strictly decoupled; when $M_s = 0$, $R_{\text{sel},s} = \text{NaN}$, status is reported as `UNDEFINED`, and confidence bounds are `UNDEFINED`.
- Operational utility is tracked separately via Coverage ($Cov_s = M_s / N_s$) and True Success Rate ($TSR_s = TS_s / N_s$). A policy that always abstains cannot be misrepresented as operationally useful because its coverage is 0.000000.
- The denominator N_s is strictly validated against `TEST_SCEN_SEEDS_PER_SCENARIO` (120). Any mismatch raises an immediate `ValueError`.

3 Remediation of Finding M4-F03: Paired Record Identity Alignment

3.1 Defect Mechanism

The previous harness executed paired comparisons in H_2 and H_3 using `zip(records, baseline_records)`. This created an implicit dependency on list ordering and offered zero validation against missing, duplicated, or permuted episodes.

3.2 Corrected Keyed Join Architecture

We implemented `align_and_pair_records` in `experiments/v9_evaluate_holdout.py`. It establishes a strict, verified bijective join on ($seed$: int, $scenario$: str):

1. **Identity Key Parsing:** Validates that `seed` is parseable as an integer and `scenario` is a non-empty string for all records.
2. **Uniqueness Verification:** Enforces that neither policy nor baseline contains duplicate $(seed, scenario)$ pairs.
3. **Cardinality and Completeness Check:** Verifies that $|D_\pi| = |D_{\text{base}}|$ and that every episode in D_π has an exact counterpart in D_{base}.
4. **Order Invariance:** Indexes baseline records into a hash table keyed by $(seed, scenario)$ and joins them in the exact evaluation order of the policy.

This ensures that discordant-pair counts (n_{10}, n_{01}) and McNemar tests are invariant to row permutations and impervious to record corruption.

Table 1: Synthetic Boundary Test Verification Results for M4-F02 and M4-F03

Test ID	Target Finding	Boundary Condition Evaluated	Theoretical Value	Result
`test_15`	M4-F02	$N_s = 120, M_s = 0, FS_s = 0$ (Full Abstention)	$p = (0.95)^{120} \approx 0.002122$	**PASS**
`test_16`	M4-F02	Invariance across $M_s \in \{0, 30, 60, 120\}$ with $FS_s = 0$	$p = 0.002122$ identical	**PASS**
`test_17a`	M4-F02	Single False Success Boundary ($N_s = 120, FS_s = 1$)	$p \approx 0.015522 \leq 0.05$	**PASS**
`test_17b`	M4-F02	Rejection on Multiple FS ($N_s = 120, FS_s = 2$)	$p \approx 0.057398 > 0.05$	**PASS**
`test_17c`	M4-F02	Severe Risk Breakdown ($N_s = 120, FS_s = 3$)	$p \approx 0.148197 > 0.05$	**PASS**
`test_18`	M4-F02	Reject non-positive N_s, total mismatch, or count asymmetry	`ValueError` raised	**PASS**
`test_19`	M4-F02	Reject outcome algebra violation ($M_s = 0, FS_s = 1$)	`ValueError` raised	**PASS**
`test_20a`	M4-F02	Berger IUT Global Pass (all 12 scenarios $FS_s \leq 1$)	$\max_s p_s \leq 0.05 \implies$ PASS	**PASS**
`test_20b`	M4-F02	Berger IUT Single Failure Rejection (11 pass, 1 fails)	$\max_s p_s > 0.05 \implies$ FAIL	**PASS**
`test_21`	M4-F02	Holm–Šidák monotonic step-down sequence for $k = 12$	$\alpha_1 = 0.004265 \ldots \alpha_{12} = 0.050$	**PASS**
`test_22`	M4-F03	Forward matching records paired cleanly	Bijective keyed match	**PASS**
`test_23`	M4-F03	Reversed baseline array order restores exact pairing	Table $(5, 5, 5, 35)$ identical	**PASS**
`test_24`	M4-F03	Reject baseline record with mismatched seed (99999)	`ValueError` raised	**PASS**
`test_25`	M4-F03	Reject baseline record with mismatched scenario	`ValueError` raised	**PASS**
`test_26`	M4-F03	Reject duplicate identity keys in policy or baseline	`ValueError` raised	**PASS**
`test_27`	M4-F03	Reject record count / length mismatch	`ValueError` raised	**PASS**
`test_28`	M4-F03	Reject missing seed, invalid string seed, or empty scenario	`ValueError` raised	**PASS**
`test_29`	M4-F03	Discordant count reconciliation ($n_{11} = 10, n_{10} = 5, n_{01} = 20, n_{00} = 65$)	$p_{\text{McNemar}} = 0.004077$	**PASS**

4 Synthetic Boundary Testing Evidence

Table 1 details the empirical results from the newly implemented synthetic unit tests in `tests/test_v9_holdout_preflight.`

5 Full Regression Audit and Immutability Verification

Following code modifications, the full repository regression suite was executed via:

```
PYTHONPATH=src uv run pytest -q
```

Result: 140 passed in 31.44s (Exit Code: 0). Zero failures, zero warnings.

5.1 Cryptographic Proof of Artifact Immutability

We verified that the calibrated policy and prior audit deliverables remain bit-for-bit unchanged:

- **Selected Policy:** `results/v9/cal_r2/selected_policy.json`
  SHA-256 = `5a89d6061b379f1b34ab47be0deef8d63180d3e4c21d18c478bc7e451f5c84e2` (**MATCH**)
- **M2R2 Calibration Report:** `results/v9/cal_r2/calibration_report.json`
  SHA-256 = `a91b9b008d7486e421be016ef3d55169992b1525a720dc55f848dd37397ef050` (**MATCH**)
- **M3 Validation Report:** `results/v9/val_r1/validation_report.json`
  SHA-256 = `c97fffa626d70eb47d34be24aa4693b047530e14cce316d296c095ec40daec4e` (**MATCH**)
- **Holdout Partitions:** `test_mix/` and `test_scen/` directories remain **UNINSTANTIATED**.

6 Conclusion and Recommendation

Findings M4-F02 and M4-F03 have been fully resolved with complete mathematical, algorithmic, and empirical rigor. The evaluation harness now adheres strictly to the registered protocol. This completes the remediation requirement.